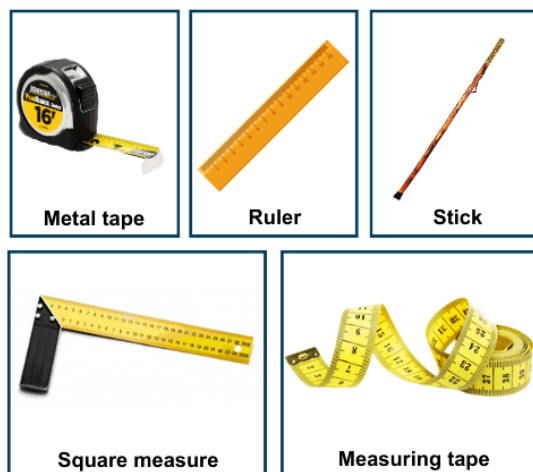


Chapter 1 – Introduction to Science

Measurements

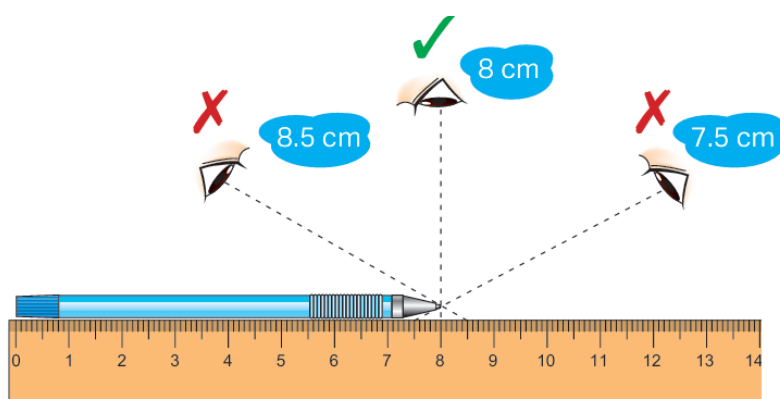
Measuring length 測量長度



Unit 單位	English	Chinese	Conversion of units
km	kilometre	千米	1000 m = 1 km
m	metre	米	-
cm	centimetre	釐米	100 cm = 1 m
mm	millimetre	毫米	1000 mm = 1 m 10 mm = 1 cm

- **Measure the length with a ruler**

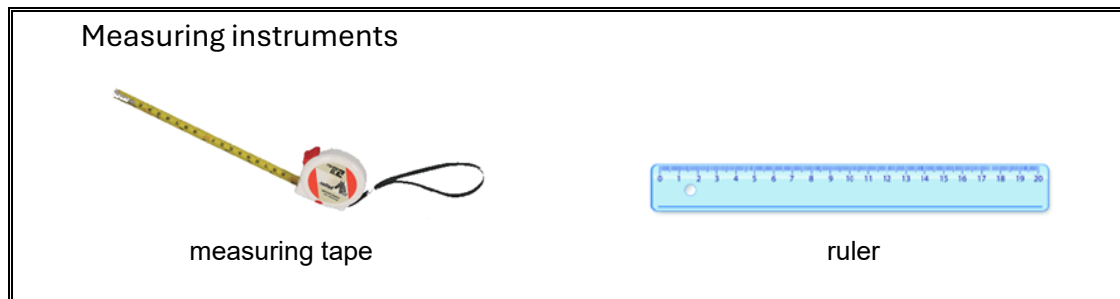
- Read the scale from vertically above 在正上方讀數



✧ Meaning of the prefix of the units

Prefix 前綴	Symbol	Numerical Value	Word Meaning
kilo-	k	1000	Thousand 千
centi-	c	0.01	Hundredth 百分之一
milli-	m	0.001	Thousandth 千分之一

In each of the following cases, choose the most appropriate measuring instrument and write the suitable unit of length for measuring the object.



(a) Height of a seedling (幼苗)



Instrument: (i) _____
Unit: (ii) _____

(b) Height of a door



Instrument: (i) _____
Unit: (ii) _____

(c) Height of a boy



Instrument: (i) _____
Unit: (ii) _____

(d) Length of the square below

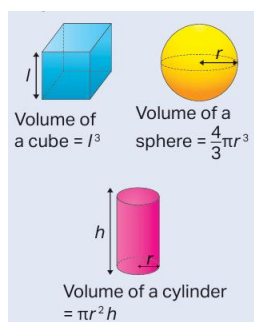


Instrument: (i) _____
Unit: (ii) _____

Measuring volume 測量體積

	Unit	English	Chinese	Conversion of units
Describe the volume of a liquid	L	litre	升	1000 mL = 1 L
	mL	millilitre	毫升	1 mL = 1 cm ³
Describe the volume	cm ³	cubic centimetre	立方釐米	1 mL = 1 cm ³
	m ³	cubic metre	立方米	1 m ³ = 1,000,000 cm ³

1. Measuring the volume of a regular object

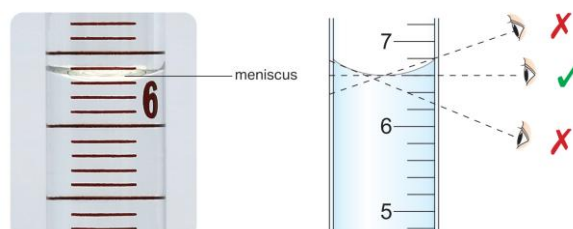


$$\text{volume} = \text{length} \times \text{width} \times \text{height}$$

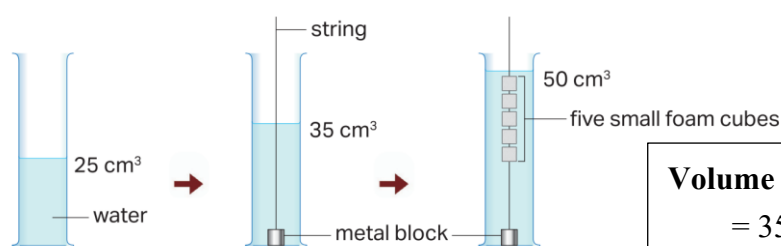


2. Measuring the volume of liquid

- Use **measuring cylinder**
- We should read at the level of the **bottom of meniscus!** 彎液面底部



3. Find the volume of a small irregular object



Volume of metal block

$$= 35 - 25 = 10 \text{ cm}^3$$

Total volume of 5 foam 泡棉 cubes

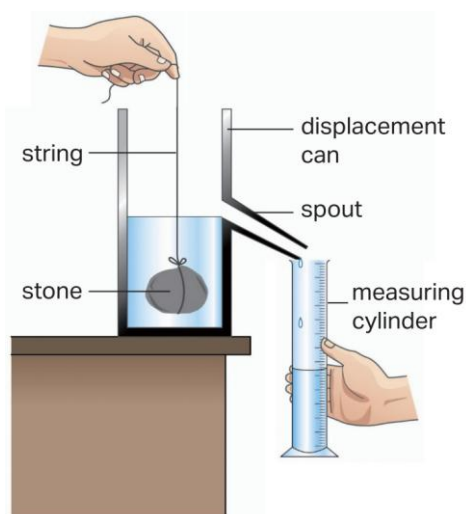
$$= 50 - 35 = 15 \text{ cm}^3$$

Volume of each foam cube

$$= 15 \div 5 = 3 \text{ cm}^3$$

Metal block is needed because foam would float on water!

4. Find the volume of a large irregular object by displacement method 排水法



- Pour water into the displacement can until a little water runs out of the spout. Remove the water collected by the measuring cylinder
- Hold the empty measuring cylinder below the spout and then slowly lower the stone into the water in the displacement can

Volume of the stone = volume of water collected (displaced)

Measure the volume of plastic balls

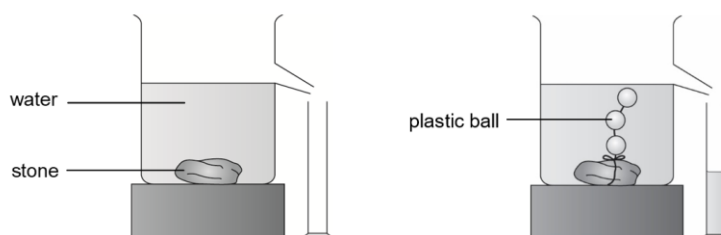


Fig a on the right shows the water level in the measuring cylinder after adding three plastic balls.

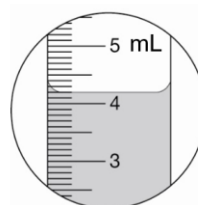


Fig a

1. How do you call the curved surface at the water level?

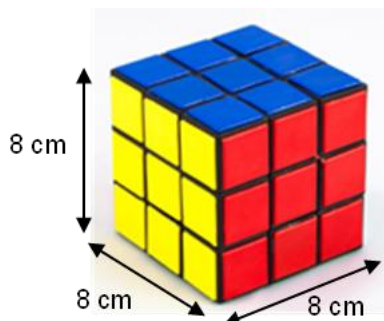
2. What is the volume reading of the measuring cylinder?

3. Calculate the volume of one plastic ball. Show your working.

Practice

1. Calculate the volume of the following regular objects.

(a) Rubik's Cube (扭計骰)



Volume

= _____

= _____

(b) Mobile phone



Volume

= _____

= _____

2. Convert each of the following values to the unit in the bracket.

(a) 375 cm³ (mL)

(b) 400 mL (cm³)

(c) 6 L (mL)

(d) 850 mL (L)

(e) 7.3 m³ (cm³)

Measuring mass 測量質量

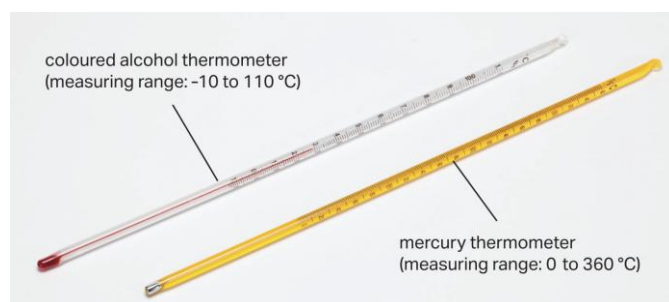
Unit	English	Chinese	Conversion of units
kg	kilogram	千克	1 kg = 1000 g
g	gram	克	-
mg	milligram	毫克	1000 mg = 1 g



triple beam balance 三桿天平
spring balance 彈簧秤
 lever balance 槓桿秤
 top pan balance 盤天平
 electronic balance 電子天平
 bathroom scale 體重計

Measuring Temperature 測量溫度

Unit	English	Chinese
°C	degree Celsius	攝氏度



thermometer 溫度計
 colored alcohol thermometer 彩色酒精溫度計
 mercury thermometer 水銀溫度計



electronic thermometer 電子體溫計

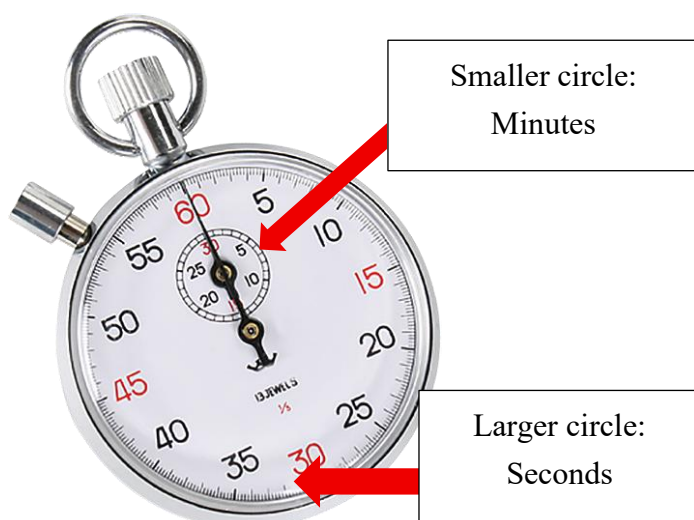
Measuring time 測量時間

Unit	English	Chinese	Conversion of units
h	hour	小時	1 h = 60 min 1 h = 3600 s
min	minute	分鐘	1 min = 60 s
s	second	秒	-



Fig. 1.42 An electronic stopwatch

Stopwatch 秒錶
watch 手錶
clock 時鐘



Answer

Measuring length 測量長度



- | | |
|------------------------|-------------|
| (a) (i) ruler | (ii) cm |
| (b) (i) measuring tape | (ii) m / cm |
| (c) (i) measuring tape | (ii) m / cm |
| (d) (i) ruler | (ii) cm |

Measuring volume 測量體積



1. Meniscus
2. 4.2 ml
3. $4.2/3 = 1.4 \text{ ml} = 1.4 \text{ cm}^3$

Practice

1. (a) $\text{Volume} = 8 \text{ cm} \times 8 \text{ cm} \times 8 \text{ cm}$
 $= 512 \text{ cm}^3$
(b) $\text{Volume} = 140 \text{ mm} \times 70 \text{ mm} \times 7 \text{ mm}$
 $= 68\,600 \text{ mm}^3 \text{ (Or } 68.6 \text{ cm}^3\text{)}$
2. (a) 375 mL
(b) 400 cm^3
(c) $6 \times 1000 \text{ mL} = \underline{6000 \text{ mL}}$
(d) Since $1 \text{ L} = 1000 \text{ mL}$, $1 \text{ mL} = (1 \div 1000) = 0.001 \text{ L}$
 $\therefore 850 \text{ mL} = 850 \times 0.001 \text{ L} = \underline{0.85 \text{ L}}$
(e) $7.3 \times 1\,000\,000 \text{ cm}^3 = \underline{7\,300\,000 \text{ cm}^3}$